

Cloud Computing & Servers

Class no:2

1. Server Definition and Types

- **Definition:** A server is a computer system or software that provides services, resources, or data to other computers (clients) over a network.
- **Types with Purpose & Example:**
 - **Web Server:** Hosts websites (e.g., Apache, Nginx).
 - **Database Server:** Stores and manages data (e.g., MySQL, Oracle DB).
 - **File Server:** Provides centralized file storage (e.g., Windows File Server).
 - **Application Server:** Runs business applications (e.g., JBoss, WebLogic).
 - **Mail Server:** Handles email communication (e.g., Microsoft Exchange).

2. AWS Service and Deployment Models

- **AWS Services:** Compute (EC2), Storage (S3), Database (RDS), Networking (VPC), AI/ML (SageMaker).
- **Deployment Models:**
 - **Public Cloud:** Services shared over the internet.
 - **Private Cloud:** Exclusive use by one organization.
 - **Hybrid Cloud:** Combination of public + private.
 - **Community Cloud:** Shared by organizations with similar needs.

3. Different Cloud Platforms & Founding Year

Cloud Platform	Founded Year
Amazon Web Services (AWS)	2006
Microsoft Azure	2010
Google Cloud Platform (GCP)	2008
IBM Cloud	2011
Oracle Cloud	2012

4. Zones and Regions

- **Region:** A geographical area containing multiple data centers (e.g., AWS Asia Pacific (Mumbai)).
- **Availability Zone (AZ):** An isolated data center within a region, designed for fault tolerance.

5. Advantages of Cloud

- Cost-effective (pay-as-you-go).
- Scalability and flexibility.
- High availability and reliability.
- Global accessibility.
- Reduced maintenance burden.

6. Limitations of Cloud

- Dependency on internet connectivity.
- Data security and privacy concerns.
- Possible downtime or outages.
- Vendor lock-in risks.
- Compliance challenges.